UC Merced Sky Port Proposal

Alan Ramirez, Edwards Castro, Jayro Ortiz-Ayala, Jason Estevez, Juan Estrada

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We are gearing up toward the era of Advanced Air Mobility (AAM), a revolutionary air transportation system that transports passengers and freight from one location to another using a vertical take-off and landing aircraft. This proposal discusses a few fundamental aspects considered for the design of a vertiport facility at UC Merced campus that seamlessly integrates AAM into campus life, and facilitates the mobility of people and goods within the CITRIS Network and beyond.

Infrastructure for Operations

In identifying the location for the vertiport, we defined its suitability based on accessibility and safety. Accessibility is defined in terms of square footage available to develop the initial infrastructure as well as room for potential expansion, and considers other features that are essential to the life and operation of the vertiport. These features were initially highlighted by the US DOT's Heliport Design and Vertiport Design Advisory Circulars and later updated by the FAA Engineering Brief 105 (2022). Besides the airside surface required for operation, other facilities including terminals, parking for employees and passengers, recharge/refueling stations, hangars, and storage are necessary. Other aspects that we considered include proximity to public transportation, the presence of bike and walking paths, and the presence of major arterial roads. Per FAA safety guidelines, parcels with an area larger than 150 by 150 feet are considered to be the standard vertiport size. We included an additional buffer of 500ft to enlarge the safety area. Considering that UC Merced is located in a rural area, at present there is no need to include further safety buffers for the presence of high-rise buildings. Based on the requirements listed above, we identified three suitable locations, notably: UC Merced main campus, Castle Air Force Base, and Merced Regional Airport. Given that two locations were off campus, we considered the time to reach the UC Merced main campus as an additional high-priority requirement for the selection of the proper location. In particular, the shorter the time to reach the main campus the higher the score for this metric. The latter was purposely included to fulfill one of the design objectives of the CITRIS Aviation Prize, i.e. ensuring at least three times faster than ground transportation to interconnect the various CITRIS campuses. Arguably, Castle Air Force Base and Merced Regional Airport are better equipped in terms of infrastructure already available (runways, hangars, etc.). This implies a reduction of costs for the development of the vertiport; however, their off-campus location adds to the total time to interconnect UC Merced to the other campuses. Based on Google Maps estimation, 15 to 20 minutes more to the total travel time are required to reach these locations from campus. The UC Merced location on the other end has a solar power farm that provides a nominal peak output of nearly $1MW$ and an annual output of nearly $1.6MWh$, which is about 9-10% of the annual electricity demand on campus. This facility is deemed extremely important since it provides access to green energy that can be used to recharge the eVTOL. Based on the consideration listed above, we considered the UC Merced location as the most viable option to further develop our proposal. A map view of the UC Merced campus which shows the selected location of the vertiport is depicted in Figure 1. Although the selected are is in proximity of a canal (see Figure 2), it is worth mentioning that the area of the vertiport is elevated with respect to the surrounding area and is not subjected to flooding. Moreover, the area selected is located nearby a parking lot, partially covered with Solar Roof, and is readily accessible from Cottonwood Lp, where we envision the presence of a bus stop and bike and/or scooter -sharing station. In Figure 3, we show a few renderings of the vertiport.

Selection of Existing Vehicle Technology

VTOLs stand out as prime candidates for future air mobility, given their distinct advantages over conventional aircraft. Notably, their vertical takeoff and landing capabilities eliminate the reliance on runways, a significant advantage in diverse environments, including remote and rugged terrains like those around UC Merced. Therefore, the initial proposed design focused on two primary AAM vehicle groups: eVTOLs and VTOLs, considered for their technological fit. When discussing the two types of AAM vehicles, the concern about environmental impact plays a key role. Here environmental impact is intended in terms of carbon and acoustic emissions.

To decide which vehicle was best for the design proposed, we conceived an evaluation matrix. Among the evaluation criteria we selected there was the range covered, the cruise speed, the unit cost, the maximum number of passengers, the amount of emission of CO_2 produced, and the acoustic emissions. In assigning the score to the different aircraft we considered the needs of our transportation network and assigned the top score to the aircraft whose performance is centered around the CITRIS network needs. For example, an aircraft whose range is significantly larger than the average distance between the campuses will see its score negatively affected.

The candidate vehicles that we initially shortlisted were: Joby Aviation's s4 eVTOL, Ody Aviation's Hybrid Electric VTOL, and the AgustaWestland AW609 VTOL. All the above vehicles fulfill the requirement of cutting the travel time to a third of the time necessary to travel by ground transportation (see Table 1).

Vehicle	UC Davis	UC Santa Cruz	UC Berkeley
Joby Aviation S4 e-VTOL	0.7 (3.35)	0.65 (3.54)	0.65 (3.54)
Ody Aviation HYBRID VTOL	0.41 (5.73)	0.38 (6.05)	0.38 (6.05)
AgustaWestland AW609 VTOL	0.45 (5.22)	0.42 (5.48)	0.42 (5.48)
Standard Road Vehicle	2.7	2.3	2.3

Table 1: Travel times in hours to get to CITRIS campuses from Merced, CA. In brackets the time reduction factor with respect to ground transportation. Travel times for the ground vehicle were taken from Google Maps. In brackets the ratio between the travel time by ground a

The AW609 and the Ody Aviation's Hybrid Electric VTOL can accommodate up to 9 passengers as opposed to the Joby Aviation eVTOL which has a passenger capacity of 5. The AW609 has the highest cruise speed compared to its competitors. With its 1150 miles range, undoubtedly the AW609 offers also the largest range; however, this is not necessary and advantage considered that the four CITRIS campuses are located at a fraction of that distance[1]. Moreover, the AW609 is propelled by conventional turboshaft engines which are less environmentally friendly when compared to the the Ody Aviation's Hybrid Electric VTOL and the Joby Aviation's s4 eVTOL. The Ody Aviation Hybrid Electric VTOL and the Joby Aviation e-VTOLs offer a similar range with electric propulsion. However, the Ody Aviation VTOL offers a 750-mile total range with hybrid electric propulsion. In terms of emissions, Joby Aviation's S4 is the best candidate since it only uses electric energy. The unit price liens in favor of the Joby Aviation too, since is the less expensive option among the three aircraft considered in this analysis. Moreover, since the Ody Aviation VTOL is hybrid, we anticipate higher maintenance and infrastructure costs as compared to its full electric competitor. While maintenance will mostly affect the carrier, infrastructure will directly affect the campus since additional safety measures and more space will be required to safely storage the fuel.

In conclusion, our analysis highlights that the best fit for connecting the four CITRIS campuses is the Joby Aviation's S4 eVTOL.

Software for Single-Vehicle Planning and Scheduling

Here we reviewed some of the most commonly adopted algorithms for single vehicle scheduling and planning. This brief review is not meant to be comprehensive nor exhaustive but highlights some essential features that are useful to make an informed decision. The choice of algorithm depends on the specific characteristics of the scheduling and planning problem, such as the size of the problem, the nature of

[1] All the CITRIS campuses are located at approximately 130 miles from UC Merced. The total loop from UC Merced-UC Davis-UC Berkley-UC Santa Cruz and back to UC Merced is approximately 300 miles.

the constraints, and the desired solution quality. Additionally, real-world applications may require a hybridization of these algorithms.

- *Greedy Algorithms.* In the context of vehicle scheduling, a greedy algorithms might prioritize tasks based on some criteria, such as distance or time windows, and schedule them sequentially. Greedy Algorithms are simple and computationally efficient but may not always result in the most optimal solution. An example of Greedy Alorithm is the Nearest Neighbor Algorithm. In this algorithm the vehicle selects the nearest unvisited location at each step.

- *Insertion Algorithms*: Insertion algorithms start with an empty schedule and iteratively insert tasks at the most appropriate positions based on certain criteria (e.g., minimizing travel distance). Compared to greedy algorithms can be more flexible but their complexity increases with the number of tasks.

- *Priority Scheduling Algorithm* is a popular algorithm used in various scheduling and planning contexts. In the context of a single vehicle, the priority scheduling algorithm assigns priorities to tasks or jobs, and the vehicle executes them in order of priority therefore an essential component of the PS Algorithm is the priority assignment.

- *Integer Linear Programming*, ILP formulates the planning and scheduling problem as an optimization problem with linear constraints and an objective function. Modifying an ILP model in response to dynamic changes can be complex, and this process may not be suitable for situations requiring rapid decision-making. ILP models may not be well-suited for scenarios requiring instantaneous responses to dynamic changes, such as sudden disruptions or emergencies.

Given its simplicity to implement and its convergence to optimal solutions for small to medium size single-vehicle problem, we believe that the Integer Linear Programming, or a slightly modified version dubbed Mixed-Integer Linear Programming, are suitable for planning and scheduling for the CITRIS network.

Architecture for Connecting to the UC CITRIS Campuses

When it comes to the architecture of the connections between the CITRIS campuses, it is essential to look at a number of potential models already successfully implemented in the air transportation sector. In particular, we focused on the following models:

- *Spoke–Hub Model* is a form of transport topology optimization in which outlying points are connected to a central "hub". The hub-and-spoke model boosts airline efficiency through optimized aircraft and crew use, enabling multiple daily trips from a central hub without returning to a home base. It reduces costs, increases profitability, and enhances passenger satisfaction by providing seamless connections to various destinations without needing separate flight bookings. The only disadvantage of using the hub and spoke traveling model is that the passenger will spend much time getting to the airport hub and catching the next flight to reach their final destination

- *Point-to-Point Model*, offers a direct connection from one destination to another in the network and eliminates the need for a central hub. In a point-to-point architecture, passengers can benefit from direct flights that significantly reduce travel time. Moreover, the flights from point to point are less interdependent as compared to the hub-spoke model, and failure in one node may not affect delays in the entire network. The point-to-point model has drawbacks, including potential inefficiency in aircraft and crew utilization, higher costs, and increased challenges during cascade failures due to the absence of a central hub for rerouting passengers.

- *Virtual Hub Model*, this model combines aspects of both hub and spoke and point-to-point models. It involves partnerships and alliances between airlines to create virtual hubs where passengers can transfer seamlessly between different carriers.

Based on our survey we believe that a point-to-point network is the best choice for the proposed intercampus connectivity because it offers the flexibility to travel directly from one campus to another without intermediate stops, minimizing the travel time for passengers and goods.

Appendix

Map_View_Campus.pdf

Figure 1: Annotated UC Merced main campus with vertiport location

detailed_view.PNG

Figure 2: Detailed view of the selected location.

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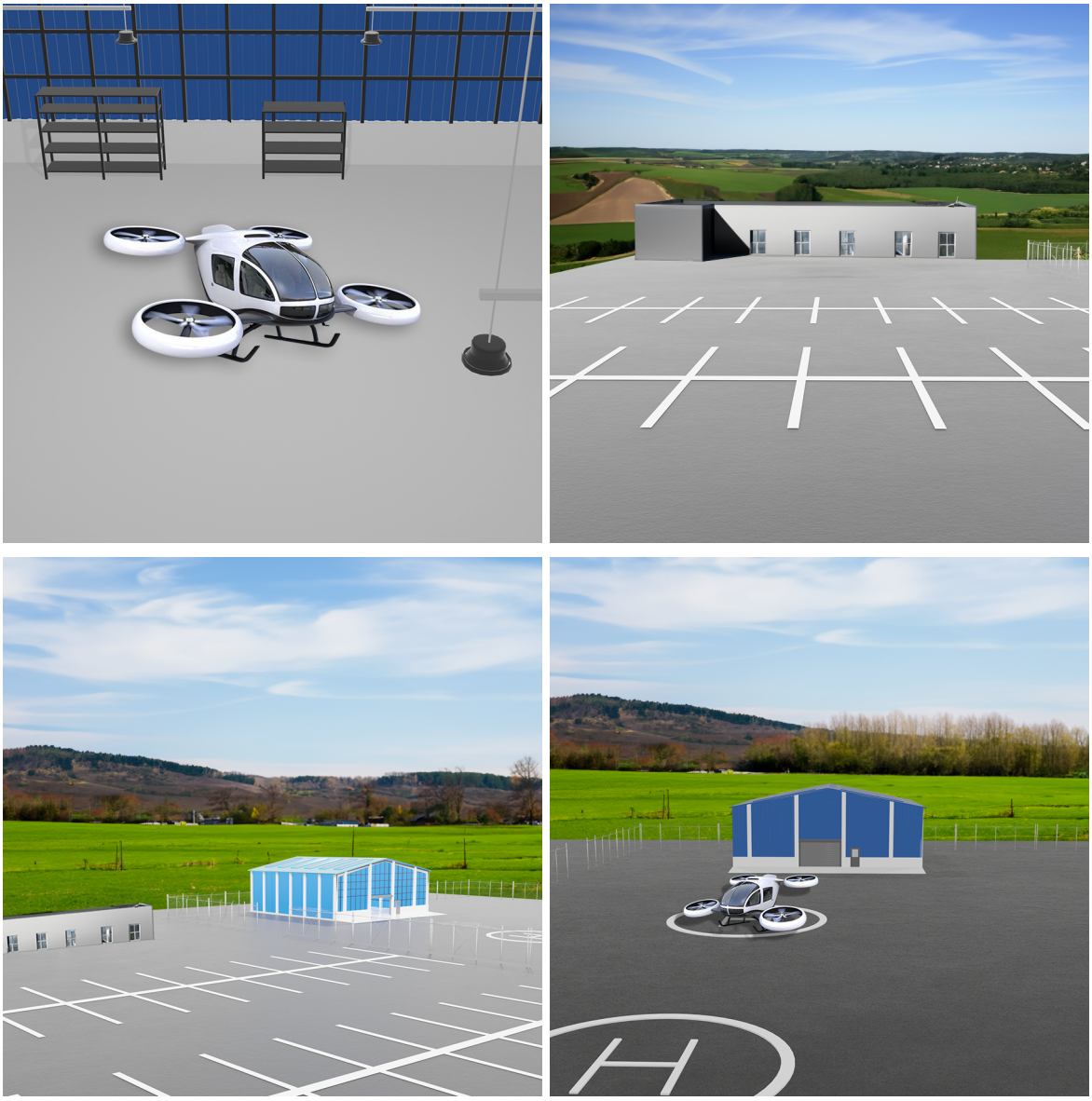

Figure 3: Renderings of the Vertiport. From left to right, and from top to bottom. Inside view of the hangar, Front View of the main building/terminal, Wide view of the vertiport, and front view of the hanger with an eVTOL on the helipad.